

**MINI PROJECT WORK ON**  
**Machine Learning Based Classification of Celestial**  
**Objects**

**SUBMITTED**  
**BY**

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*under the supervision of*

**PRITHA DAS**



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MAY 2026  
Howrah

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# Abstract

In recent years, the field of astronomy has experienced a massive growth in data generation due to advanced telescopes and sky surveys. These instruments continuously collect information about celestial objects such as galaxies, stars, and quasars. The classification of these objects is a crucial task because it helps scientists understand the structure and evolution of the universe.

Traditionally, classification was performed manually by astronomers through observation and analysis. However, this approach is no longer practical due to the enormous volume of data. Therefore, automated methods using machine learning have become essential.

In this project, a Random Forest machine learning model is used to classify celestial objects based on numerical features such as redshift, photometric measurements (u, g, r, i, z), and positional attributes (alpha and delta). The dataset used in this study consists of labeled observations of celestial objects, where each object is categorized as a galaxy, star, or quasar (QSO).

The data is preprocessed by converting categorical labels into numerical form and splitting the dataset into training and testing sets. The Random Forest model is then trained on the training data and evaluated using test data.

The performance of the model is measured using accuracy and confusion matrix. The model achieved an accuracy of approximately **97.87%**, indicating a high level of classification performance. The confusion matrix further shows that most predictions are correct, with only minor misclassifications occurring due to similarities between certain classes.

This project demonstrates the effectiveness of machine learning in solving real-world problems in astronomy. It highlights how data-driven approaches can significantly improve efficiency, reduce human effort, and provide accurate and reliable results in large-scale scientific data analysis.

# 1 Introduction

Astronomy is the scientific study of celestial objects such as stars, galaxies, and quasars, as well as the physical processes that govern the universe. Over the past few decades, advancements in observational technologies, including space telescopes and large-scale sky surveys, have led to an exponential increase in the amount of astronomical data collected. Projects such as the Sloan Digital Sky Survey (SDSS) generate massive datasets containing millions of observations of celestial objects.

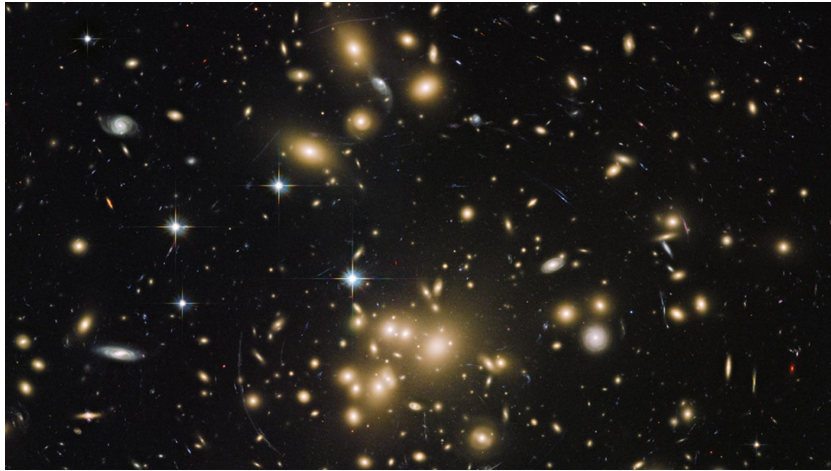


Figure 1: Astronomical observations of galaxies and stars

While this abundance of data provides valuable insights into the universe, it also presents a significant challenge. Manually analyzing and classifying such large datasets is not only time-consuming but also requires expert knowledge and is prone to human error. As the volume of data continues to grow, traditional methods become inefficient.

Machine learning, a branch of artificial intelligence, offers a powerful solution to this problem. It enables computers to learn patterns from data and make predictions without being explicitly programmed. In the context of astronomy, machine learning can be used to automatically classify celestial objects based on their observed features, such as brightness, position, and distance.

In this project, a machine learning approach is used to classify celestial objects into three categories: galaxy, star, and quasar (QSO). These objects

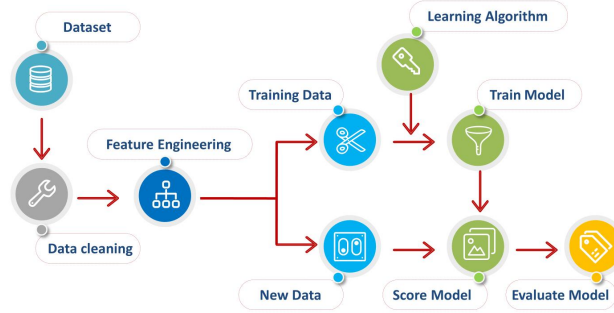


Figure 2: Machine learning workflow for classification

differ in their physical properties and behavior. Galaxies are large systems of stars bound together by gravity, stars are luminous objects that produce energy through nuclear fusion, and quasars are extremely bright objects powered by supermassive black holes.

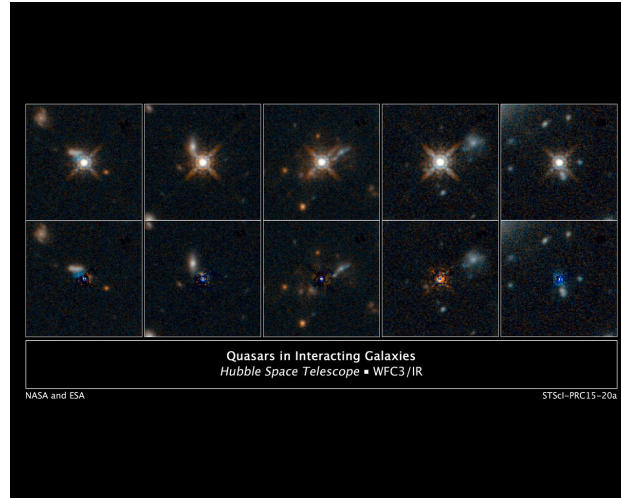


Figure 3: Types of celestial objects: Galaxy, Star, and Quasar

The classification is performed using a Random Forest algorithm, which is an ensemble learning method that combines multiple decision trees to improve accuracy and reduce errors. The model is trained on a dataset containing numerical features such as redshift, photometric measurements, and positional coordinates.

The primary aim of this project is to develop an efficient and accurate

classification model that can automatically identify the type of celestial object based on its features. By doing so, the project demonstrates how machine learning can be applied to real-world scientific problems, particularly in handling large-scale astronomical data.

This study not only highlights the importance of automation in modern astronomy but also shows how mathematical models and computational techniques can be integrated to achieve reliable and efficient results.

## 2 Problem Statement

The rapid advancement of astronomical observation technologies has resulted in the generation of extremely large datasets containing information about celestial objects such as galaxies, stars, and quasars. These datasets include various numerical features such as brightness, position, and distance measurements.

Traditionally, astronomers classified these objects manually by analyzing observational data. However, with the increasing volume of data, manual classification has become impractical. It requires significant time, expertise, and effort, and is also prone to human error.

In addition, different celestial objects often share similar characteristics, making it difficult to distinguish between them using traditional methods. This creates a need for an automated system that can accurately classify objects based on their features.

Therefore, the main problem addressed in this project is:

To develop an automated and accurate system for classifying celestial objects into galaxy, star, and quasar categories using machine learning techniques.

## 3 Objectives

The objectives of this project are designed to address the problem effectively and systematically.

The first objective is to understand and analyze the dataset. This involves studying the structure of the data, identifying the features, and understanding their significance in astronomy.

The second objective is to preprocess the data. Since machine learning models require numerical input, categorical labels such as Galaxy, Star, and QSO must be converted into numerical form.

The third objective is to develop a machine learning model using the Random Forest algorithm, which is known for its high accuracy and ability to handle complex datasets.

The fourth objective is to train and test the model using separate datasets to ensure that it can generalize well to new data.

The final objective is to evaluate the performance of the model using metrics such as accuracy and confusion matrix.

## 4 Scope of the Project

The scope of this project defines the boundaries and limitations of the work carried out.

This project focuses on the classification of celestial objects using numerical dataset features only. It does not involve image-based classification or deep learning techniques.

The classification is limited to three categories:

- Galaxy
- Star
- Quasar (QSO)

The implementation is carried out using Python and machine learning libraries such as Pandas and Scikit-learn.

While the model achieves high accuracy, its performance depends on the dataset used and may vary when applied to different datasets.

Despite these limitations, the project successfully demonstrates how machine learning can be used to automate classification tasks in astronomy and handle large datasets efficiently.

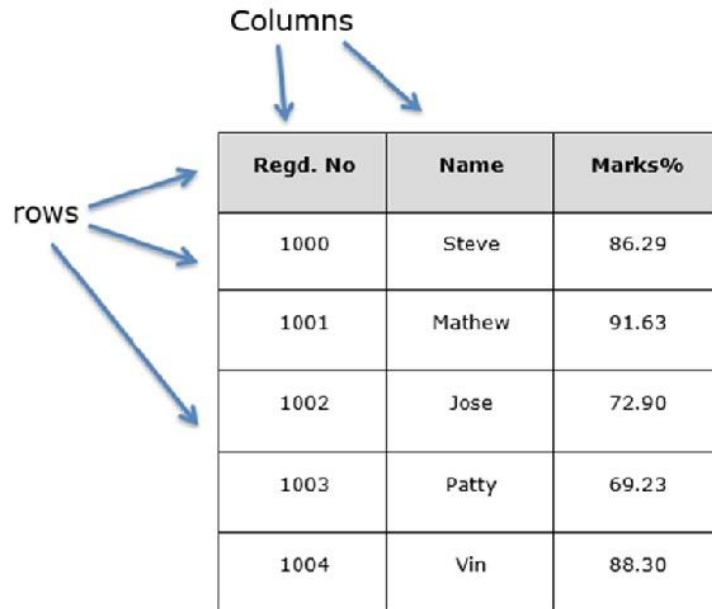
## 5 Dataset Description

### 5.1 Overview of the Dataset

The dataset used in this project consists of numerical observations of celestial objects collected through astronomical surveys. Each row in the dataset represents a single object, while each column represents a feature that describes a specific property of that object.

These features include measurements related to the position, brightness, and distance of the objects. The dataset is labeled, meaning that each object is already classified as a galaxy, star, or quasar (QSO). This makes it suitable for supervised machine learning.

### 5.2 Structure of the Dataset



The diagram illustrates the structure of a dataset table. A table with three columns and six rows is shown. The columns are labeled 'Regd. No', 'Name', and 'Marks%'. The rows contain data for five objects, with the first row being the header. Arrows point from the word 'Columns' to the column headers, and from the word 'rows' to the data rows.

| Regd. No | Name   | Marks% |
|----------|--------|--------|
| 1000     | Steve  | 86.29  |
| 1001     | Mathew | 91.63  |
| 1002     | Jose   | 72.90  |
| 1003     | Patty  | 69.23  |
| 1004     | Vin    | 88.30  |

Figure 4: Sample structure of dataset showing rows and columns

The dataset is organized in tabular form where:

- Rows represent individual celestial objects



- Columns represent features such as brightness, position, and distance

### 5.3 Input Features

| Feature       | Description              | Purpose                 |
|---------------|--------------------------|-------------------------|
| alpha         | Right ascension          | Position in sky         |
| delta         | Declination              | Vertical position       |
| u, g, r, i, z | Light intensity values   | Identify object type    |
| redshift      | Distance indicator       | Distinguish far objects |
| plate         | Observation plate number | Data reference          |
| MJD           | Observation date         | Time reference          |
| fiber_ID      | Fiber identifier         | Instrument detail       |

Table 1: Input features used in the dataset

### 5.4 Explanation of Key Features

**Redshift:** Redshift measures how much the wavelength of light from an object has shifted due to the expansion of the universe. Objects with higher redshift values are generally farther away. This feature is important because galaxies and quasars are usually located at large distances compared to stars.

**Photometric Bands (u, g, r, i, z):** These features represent brightness of objects in different wavelength ranges. Each celestial object emits light differently across these bands, which helps the model distinguish between object types.

**Alpha and Delta:** These represent the coordinates of objects in the sky and help in identifying their spatial location.

### 5.5 Target Variable

The class labels are converted into numerical values because machine learning algorithms require numerical input.

### 5.6 Data Distribution

The dataset contains:

| Class Name | Encoded Value |
|------------|---------------|
| Galaxy     | 0             |
| Star       | 1             |
| QSO        | 2             |

Table 2: Target variable encoding

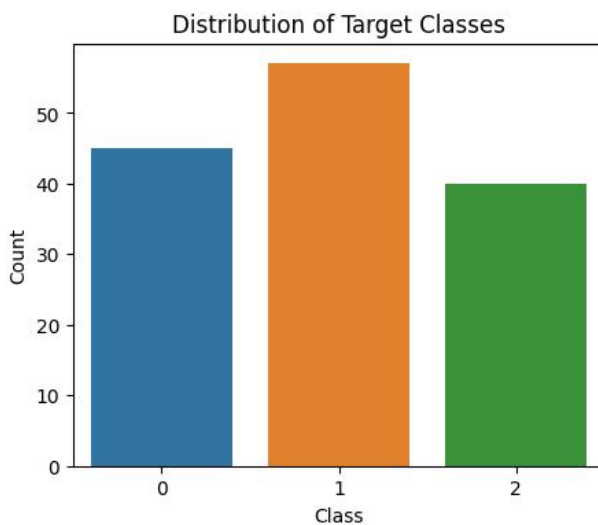


Figure 5: Distribution of Galaxy, Star, and QSO classes

- Galaxy: 59,445 samples
- Star: 21,594 samples
- QSO: 18,961 samples

This indicates that the dataset is slightly imbalanced, with galaxies having the highest number of samples.

## 5.7 Importance of the Dataset

The dataset plays a crucial role in this project. A well-structured dataset with relevant features allows the machine learning model to learn meaningful patterns and achieve high accuracy.

In this project, the dataset provides sufficient information to distinguish between different celestial objects, which contributes to the overall performance of the model.

## 6 Theoretical Background

### 6.1 Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables systems to learn patterns from data and make predictions without being explicitly programmed. Instead of following predefined rules, machine learning models improve their performance by learning from examples.

In this project, machine learning is used to classify celestial objects into different categories based on their features.

### 6.2 Types of Machine Learning

Machine learning can be classified into three main types:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

This project uses **supervised learning**, where the dataset contains both input features and corresponding output labels.

### 6.3 Classification Problem

Classification is a supervised learning task where the goal is to assign a label to input data.

Mathematically, it can be represented as:

$$f(x) = y$$

where:

- $x$  represents input features

- $y$  represents output class

In this project:

- Input: redshift, brightness, position
- Output: Galaxy, Star, QSO

## 6.4 Feature Space

Each object in the dataset is represented in a multi-dimensional feature space.

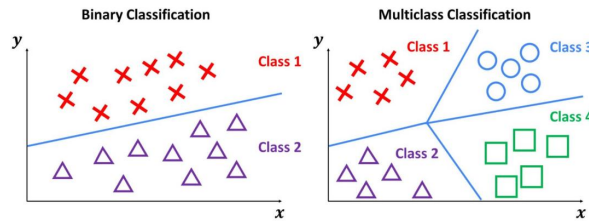


Figure 6: Feature space showing classification boundaries

In this space, the model learns boundaries that separate different classes.

## 6.5 Decision Tree

A decision tree is a basic classification model that splits data into branches based on conditions.

Each node represents a decision, and each branch represents an outcome. The final nodes (leaves) represent class labels.

However, a single decision tree may overfit the data and produce inaccurate results.

## 6.6 Random Forest Algorithm

Random Forest is an ensemble learning method that combines multiple decision trees to improve performance.

**Working of Random Forest:**

1. Multiple decision trees are created

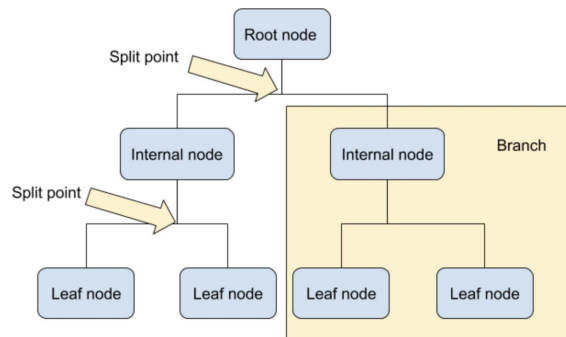


Figure 7: Decision tree structure

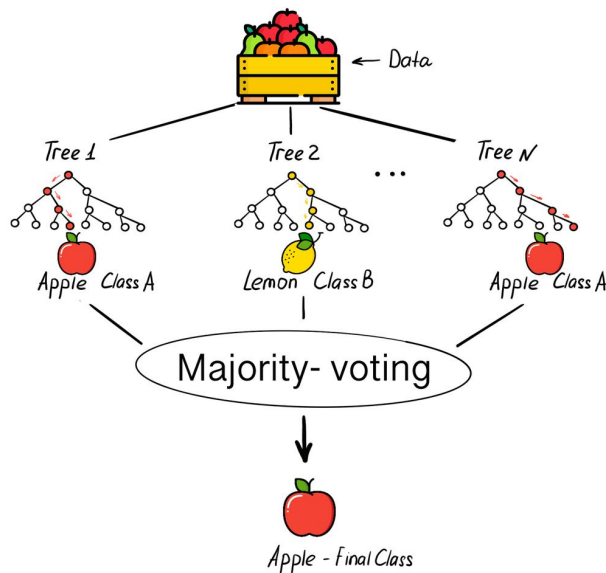


Figure 8: Random Forest model combining multiple trees

2. Each tree is trained on a subset of data
3. Each tree makes a prediction
4. Final prediction is obtained by majority voting

#### Advantages:

- Reduces overfitting

- Handles large datasets
- Provides high accuracy

## 6.7 Overfitting and Underfitting

**Overfitting:** When a model learns the training data too well and fails on new data.

**Underfitting:** When a model is too simple to capture patterns in the data.

Random Forest helps reduce overfitting by combining multiple models.

## 6.8 Model Evaluation Metrics

**Accuracy:**

$$Accuracy = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

In this project, the accuracy achieved is **97.87%**.

**Confusion Matrix:**

| Confusion Matrix       |                       |                       |
|------------------------|-----------------------|-----------------------|
|                        | Actually Positive (1) | Actually Negative (0) |
| Predicted Positive (1) | True Positives (TPs)  | False Positives (FPs) |
| Predicted Negative (0) | False Negatives (FNs) | True Negatives (TNs)  |

Figure 9: Confusion matrix representation

A confusion matrix provides a detailed breakdown of correct and incorrect predictions.

## 6.9 Importance of Theory

The theoretical concepts discussed in this chapter form the foundation of this project. Understanding machine learning, classification, and Random Forest is essential for building an effective model and interpreting the results correctly.

## 7 Methodology

### 7.1 Overview

The methodology of this project follows a structured machine learning pipeline, starting from data collection and ending with model evaluation. Each step is carefully designed to ensure that the model learns meaningful patterns and produces accurate predictions.

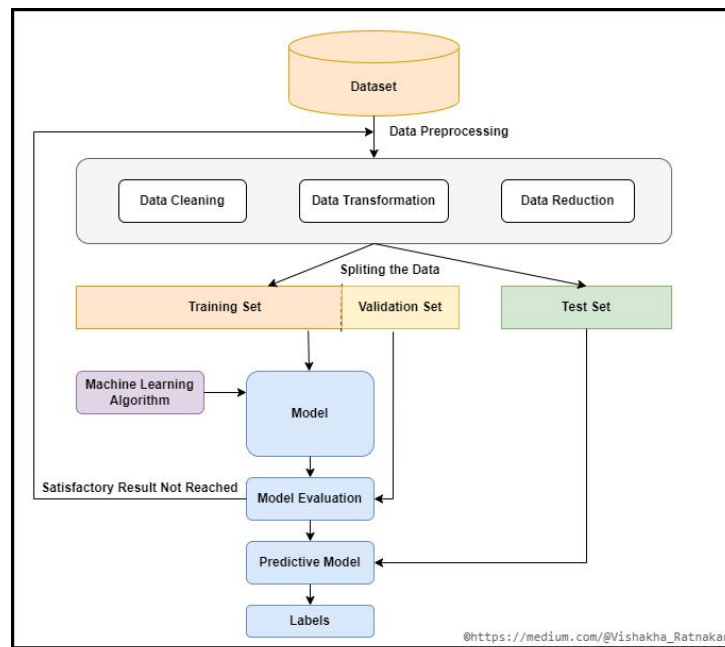


Figure 10: Machine learning workflow used in this project

## 7.2 Step 1: Data Collection

The first step involves obtaining the dataset containing information about celestial objects. The dataset is stored in CSV format and includes features such as redshift, brightness, and positional coordinates.

**Purpose:** To provide the raw data required for building the model.

## 7.3 Step 2: Data Loading

The dataset is loaded into Python using the Pandas library.

**Purpose:** To convert raw data into a structured format for analysis.

## 7.4 Step 3: Data Exploration

The dataset is explored to understand its structure and contents.

**Activities performed:**

- Viewing first few rows
- Checking data types
- Understanding feature distribution

**Purpose:** To ensure the dataset is suitable for machine learning.

## 7.5 Step 4: Data Preprocessing

The class labels are converted into numerical values.

- Galaxy  $\rightarrow 0$
- Star  $\rightarrow 1$
- QSO  $\rightarrow 2$

**Purpose:** Machine learning algorithms require numerical input.



## 7.6 Step 5: Feature Selection

The dataset is divided into:

- Input features ( $X$ )
- Target variable ( $y$ )

**Purpose:** To separate independent variables from the output.

## 7.7 Step 6: Train-Test Split

The dataset is divided into training and testing sets.

- 80% training data
- 20% testing data

**Purpose:** To evaluate model performance on unseen data.

## 7.8 Step 7: Model Training

The Random Forest algorithm is used to train the model using the training dataset.

**Purpose:** To allow the model to learn patterns from the data.

## 7.9 Step 8: Prediction

The trained model is used to predict the class of test data.

**Purpose:** To test the model's ability to generalize.

## 7.10 Step 9: Model Evaluation

The model is evaluated using:

- Accuracy
- Confusion Matrix

**Purpose:** To measure the performance and reliability of the model.

## 7.11 Summary

The methodology ensures a complete pipeline from data preparation to model evaluation. Each step contributes to achieving high accuracy and reliable classification results.

# 8 Implementation

## 8.1 Overview

The implementation of this project was carried out using the Python programming language. Python is widely used in machine learning and data science due to its simplicity and availability of powerful libraries.

In this project, Python was used to load the dataset, preprocess the data, train the machine learning model, and evaluate the results.

## 8.2 Development Environment

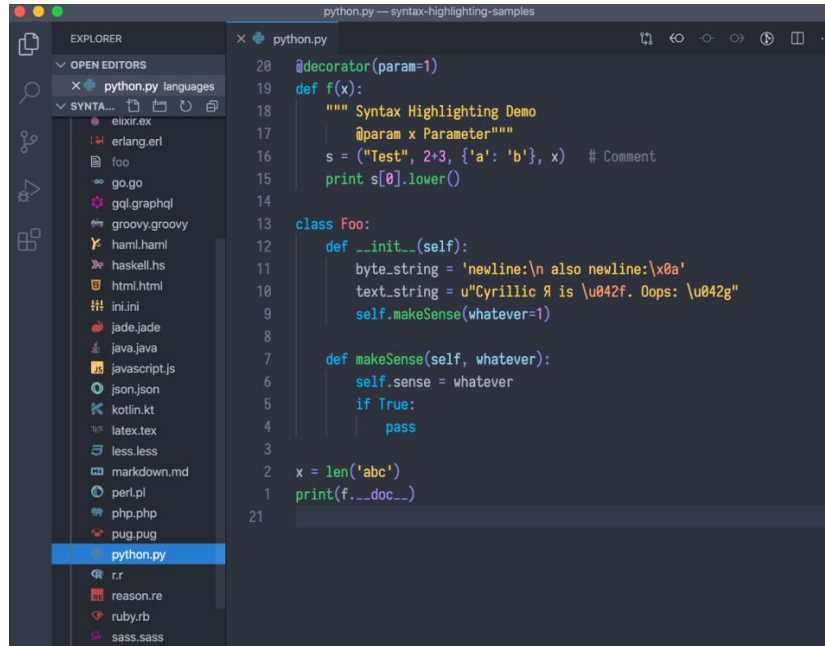


Figure 11: Python development environment used for implementation

The project was executed using a Python environment such as VS Code or command prompt, where scripts were run and outputs were observed.

### 8.3 Libraries Used

Several Python libraries were used in this project, each serving a specific purpose.

**1. Pandas:** Pandas is used for data manipulation and analysis. It helps in loading the dataset, viewing data, and preprocessing.

**2. NumPy:** NumPy is used for numerical computations and efficient handling of arrays.

**3. Scikit-learn:** Scikit-learn provides machine learning algorithms and tools such as:

- Random Forest Classifier
- Train-Test Split
- Accuracy Score

**4. Matplotlib:** Matplotlib is used for visualization, especially for plotting the confusion matrix.

### 8.4 Implementation Workflow

The implementation follows a sequence of steps:

1. Load the dataset using Pandas
2. Preprocess the data (encode class labels)
3. Separate features (X) and target variable (y)
4. Split data into training and testing sets
5. Train the Random Forest model
6. Predict using test data
7. Evaluate using accuracy and confusion matrix

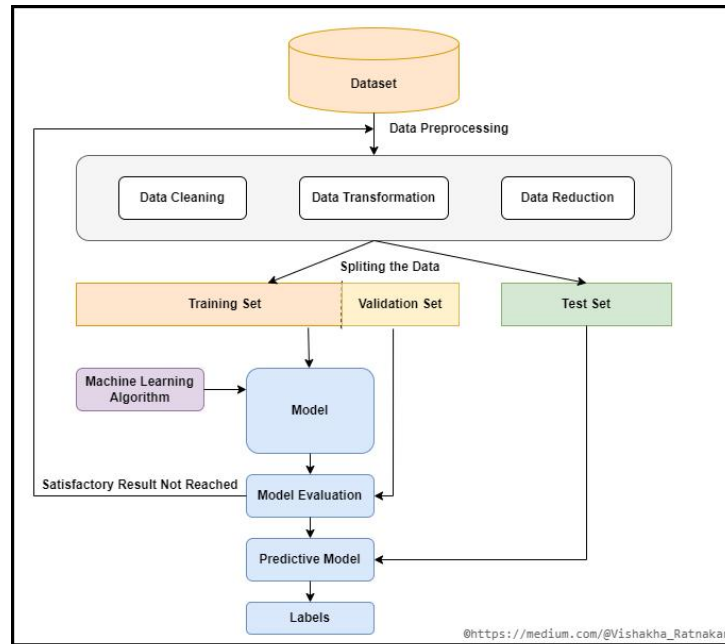


Figure 12: Implementation workflow of the project

## 8.5 Why Random Forest Was Used

The Random Forest algorithm was chosen because:

- It handles large datasets efficiently
- It reduces overfitting
- It provides high accuracy
- It works well with numerical data

## 8.6 Challenges Faced

During implementation, some challenges were encountered:

- Installing required libraries (e.g., Pandas)
- Handling file path issues

- Ensuring correct data format

These challenges were resolved by installing missing packages and verifying dataset paths.

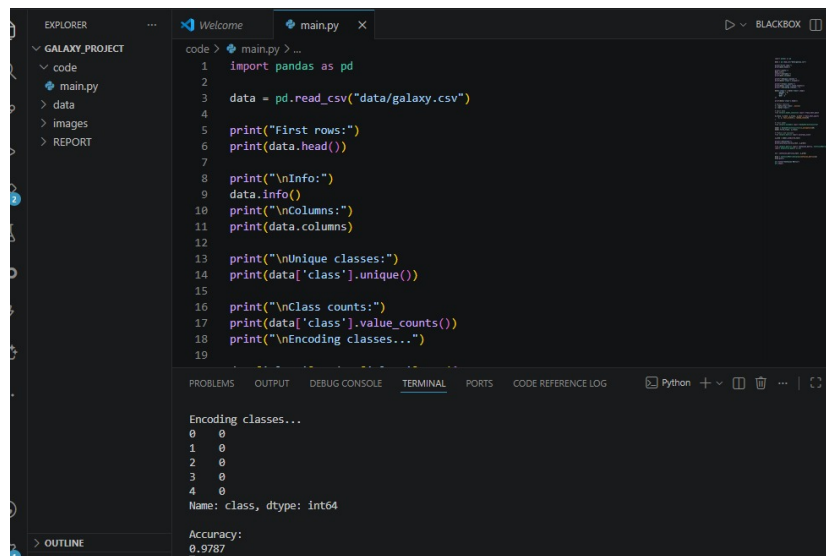
## 8.7 Summary

The implementation successfully transforms theoretical concepts into a working system. The use of Python and machine learning libraries allowed efficient data processing, model training, and evaluation.

# 9 Results

## 9.1 Model Accuracy

After training the Random Forest model, it was tested on unseen data (test dataset). The predictions made by the model were compared with the actual values to evaluate its performance.



```
code > main.py > ...
1 import pandas as pd
2
3 data = pd.read_csv("data/galaxy.csv")
4
5 print("First rows:")
6 print(data.head())
7
8 print("\nInfo:")
9 data.info()
10 print("\nColumns:")
11 print(data.columns)
12
13 print("\nUnique classes:")
14 print(data['class'].unique())
15
16 print("\nClass counts:")
17 print(data['class'].value_counts())
18 print("\nEncoding classes...")
19
```

```
Encoding classes...
0 0
1 0
2 0
3 0
4 0
Name: class, dtype: int64

Accuracy:
0.9787
```

Figure 13: Model accuracy output from Python

The model achieved an accuracy of **97.87%**.

**Explanation:**

Accuracy represents the proportion of correct predictions made by the model. An accuracy of 97.87% means that out of every 100 celestial objects, approximately 98 objects are classified correctly.

## 9.2 Interpretation of Results

The high accuracy indicates that:

- The dataset contains meaningful features
- The Random Forest algorithm is effective
- The model has successfully learned patterns from the data

## 9.3 Limitations of Accuracy

Although the accuracy is high, it does not provide complete information about the model's performance. Some classes may still be misclassified. Therefore, a more detailed analysis is required using a confusion matrix.

## 9.4 Confusion Matrix

The confusion matrix provides a detailed breakdown of the model's performance by comparing actual values with predicted values.

## 9.5 Detailed Analysis

From the confusion matrix:

- Galaxy (Class 0): Most galaxies are correctly classified, with very few errors.
- Star (Class 1): Stars are classified with almost perfect accuracy.
- QSO (Class 2): Some quasars are misclassified as galaxies.

### **Reason for Errors:**

Misclassification occurs because some celestial objects share similar features, making them difficult to distinguish.

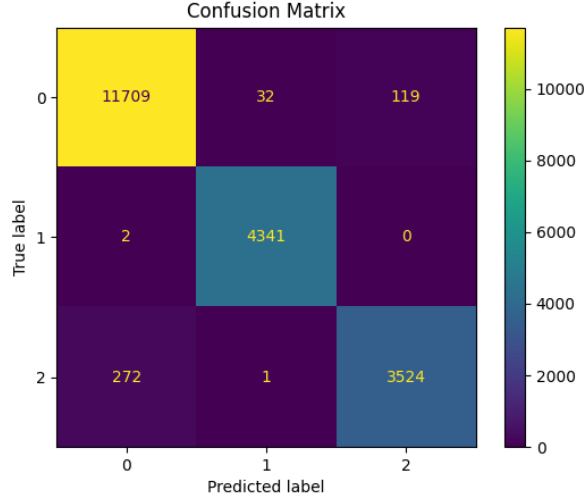


Figure 14: Confusion matrix of model predictions

## 9.6 Overall Performance

The results show that the model performs very well across all classes. The high accuracy and strong confusion matrix results indicate that the model is reliable and suitable for classification tasks.

# 10 Discussion and Insights

## 10.1 Overview of Model Performance

The results obtained from the model demonstrate that the Random Forest algorithm performs very effectively for the classification of celestial objects. With an accuracy of approximately **97.87%**, the model shows a high level of reliability in predicting the correct class of objects.

This high performance indicates that the dataset contains meaningful and discriminative features, and the model has successfully learned the relationships between these features and the target classes.

## 10.2 Interpretation of Results

The confusion matrix provides deeper insight into the model's behavior.

- The model performs exceptionally well in classifying stars, with almost no misclassification.
- Galaxies are also classified with high accuracy, with only a small number of errors.
- Quasars (QSO) show slightly higher misclassification, particularly being confused with galaxies.

**Reason:**

Certain celestial objects share similar physical characteristics such as brightness and redshift values. Due to this similarity, the model may find it difficult to distinguish between them in some cases.

## 10.3 Feature Importance

Different features contribute differently to the classification process.

**Key observations:**

- Redshift plays a major role in distinguishing distant objects
- Photometric values help identify brightness patterns
- Positional features provide additional context

## 10.4 Strengths of the Model

The model has several strengths:

- High accuracy (97.87%)
- Robust performance across all classes
- Ability to handle large datasets
- Reduced overfitting due to ensemble learning



## 10.5 Limitations of the Model

Despite strong performance, some limitations exist:

- Slight confusion between Galaxy and QSO classes
- Dependence on dataset quality
- Limited to numerical features only

## 10.6 Real-World Comparison

In real-world astronomy, data is often complex and noisy. The performance of this model reflects real-world conditions where high accuracy is achievable but small errors are expected.

## 10.7 Key Insights

This project provides several important insights:

- Machine learning can automate classification tasks efficiently
- Feature selection plays a critical role in model performance
- Ensemble methods like Random Forest improve accuracy
- Real-world data always contains some level of uncertainty

## 10.8 Conclusion of Discussion

Overall, the model demonstrates strong performance with minimal errors. The results confirm that the approach used in this project is effective and suitable for real-world applications in astronomy.

# 11 Applications and Real-World Impact

## 11.1 Application in Astronomy Research

Astronomy research involves studying celestial objects to understand the structure, origin, and evolution of the universe. Modern astronomical surveys generate massive datasets containing millions of observations.



Figure 15: Astronomical data analysis in research

The machine learning model developed in this project can automatically classify celestial objects into galaxies, stars, and quasars.

**Impact:**

- Reduces manual effort
- Speeds up data analysis
- Improves accuracy in classification

## 11.2 Application in Space Exploration

Space exploration missions collect large volumes of data from satellites and telescopes.

The model can be used to analyze this data and classify objects in real time.

**Impact:**

- Helps identify important objects
- Supports scientific decision-making
- Enables faster discoveries



Figure 16: Data collection in space missions

### 11.3 Application in Automated Systems

Machine learning models are widely used in automated data processing systems.

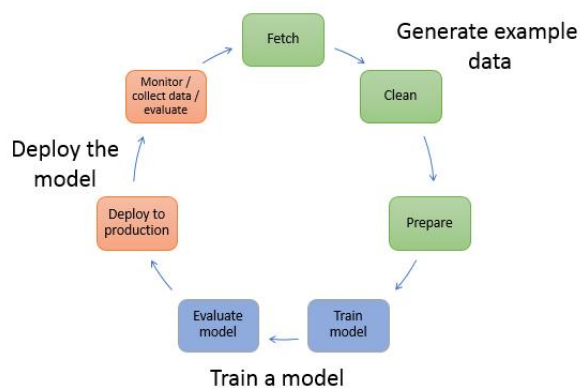


Figure 17: Automated data processing system

The model developed in this project acts as an automated classification system.

**Impact:**

- Eliminates manual classification
- Ensures consistency
- Increases efficiency

## 11.4 Importance of Automation

In earlier times, astronomers manually classified objects. However, with the increasing volume of data, automation has become necessary.

This project demonstrates how machine learning can be used to automate classification tasks effectively.

## 11.5 Broader Applications

The techniques used in this project can also be applied in other fields such as:

- Medical diagnosis
- Fraud detection
- Image recognition

## 11.6 Summary

This project highlights the importance of machine learning in solving real-world problems and demonstrates its practical applications in astronomy and beyond.

# 12 Conclusion

The objective of this project was to develop a machine learning model capable of automatically classifying celestial objects into three categories: galaxy, star, and quasar (QSO). This objective was successfully achieved by implementing a Random Forest classifier on a structured astronomical dataset.

The project began with understanding the dataset and identifying important features such as redshift, photometric values, and positional coordinates. These features played a crucial role in distinguishing between different types of celestial objects. Proper data preprocessing, including encoding of class labels and splitting of the dataset, ensured that the model could be trained effectively.

The Random Forest algorithm was chosen due to its ability to handle large datasets, reduce overfitting, and provide high accuracy. After training and testing the model, an accuracy of **97.87%** was achieved. This high accuracy indicates that the model has successfully learned patterns from the dataset and can make reliable predictions.

Further analysis using the confusion matrix showed that most predictions were correct, with only minor misclassifications occurring between certain classes, particularly between galaxies and quasars. These errors are expected in real-world data due to overlapping characteristics.

## 12.1 What Was Achieved

The following outcomes were successfully achieved:

- Developed a complete machine learning pipeline
- Implemented a Random Forest classification model
- Achieved high accuracy in classification
- Evaluated performance using confusion matrix
- Demonstrated real-world applicability

## 12.2 Final Insight

This project demonstrates that machine learning can effectively solve real-world problems in astronomy. It highlights the importance of data-driven approaches and shows how automation can improve efficiency, accuracy, and scalability in large-scale data analysis.

## 13 Future Scope

The current project demonstrates the successful application of machine learning in classifying celestial objects using numerical data. While the results are highly accurate, there are several ways in which this work can be further improved and extended in the future.

One possible improvement is the use of deep learning techniques. Deep learning models, such as Convolutional Neural Networks (CNNs), can analyze image data and capture more complex patterns compared to traditional machine learning models.

Another improvement is the use of larger and more diverse datasets. Training the model on data from multiple astronomical surveys can improve its robustness and generalization ability.

The project can also be extended to support real-time data processing. Integrating the model with live data from telescopes would allow immediate classification of newly observed celestial objects.

Further enhancements can include feature engineering and selection to improve model performance. Advanced techniques such as dimensionality reduction can also be applied.

Additionally, other machine learning algorithms such as Support Vector Machines (SVM) and Gradient Boosting can be explored and compared with Random Forest.

Finally, the model can be integrated into a user-friendly interface or application, making it accessible for researchers and students.

## 14 References

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